

LAB EXPERIMENT REPORT

Experiment 4

Data Modelling and Form Design for a Business Application in Zoho Creator

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Application Built: Appointment_Booking_Form (Zoho Creator)

1. Aim

To perform data modelling for a hospital appointment booking system and to design and build the corresponding data entry form, named Appointment_Booking_Form, in Zoho Creator, complete with the required fields, field-level validations, and a report view for viewing submitted records.

2. Software / Tools Required

- Zoho Creator (low-code application development platform) — trial account
- Web browser with internet connectivity
- A draw.io / ER-modelling tool for designing the Entity Relationship (ER) diagram

3. Theory

Data modelling is the process of identifying the entities involved in a business process, the attributes that describe each entity, and the relationships that connect those entities. A good data model forms the foundation for any application, because it decides how data will be stored, related and retrieved later. In this experiment, the business process chosen is hospital appointment booking, where a patient registers personal and contact details, is associated with a department and one or more existing medical conditions, and books a preferred appointment slot with a consent declaration.

Zoho Creator is a low-code application development platform that allows a data model to be translated directly into a working form and an accompanying report/table view, without writing extensive back-end code. Each entity of the data model becomes a form (or a linked form), each attribute becomes a field of the appropriate field type (text, number, date, dropdown, etc.), and each relationship between entities is implemented using lookup fields. Once the form is published, every submission is automatically stored as a record and can be viewed, filtered and reported through the Report view.

The general workflow followed for this experiment is: (i) identify the entities and attributes relevant to the appointment booking process, (ii) draw the ER diagram showing primary keys, foreign keys and cardinalities, (iii) translate the model into a Zoho Creator form by grouping related attributes into sections and choosing the appropriate field type for each attribute, (iv) apply field-level validations so that only clean, correctly formatted data can be submitted, and (v) publish the form and verify it by submitting sample records and checking the auto-generated report view.

3.1 Procedure

- Signed in to Zoho Creator and created a new application named Appointment_Booking_Form.
- Designed the ER diagram for the PATIENT, APPOINTMENT, DEPARTMENT, MEDICAL_CONDITION and PATIENT_CONDITION entities.
- Opened the Form Builder and added the 13 fields required, grouping them into the Patient Details, Contact Details, Medical Details and Scheduling Consent sections.
- Configured field properties — marked mandatory fields, set the phone field's country code, and set the date field to disallow past dates.
- Switched to Live Preview to verify the layout and field arrangement of the published form.
- Submitted three sample patient records through the live form to test end-to-end functionality.

- Opened the auto-generated Appointment_Booking_Form Report and verified that all submitted records appeared correctly with every field populated as expected.

4. Entity Relationship (ER) Diagram

Before building the form, the underlying data model was designed as an ER diagram. Five entities were identified: PATIENT, APPOINTMENT, DEPARTMENT, MEDICAL_CONDITION and the associative entity PATIENT_CONDITION (used to resolve the many-to-many relationship between patients and medical conditions). The diagram below shows the primary keys (PK), foreign keys (FK) and the cardinality of each relationship.

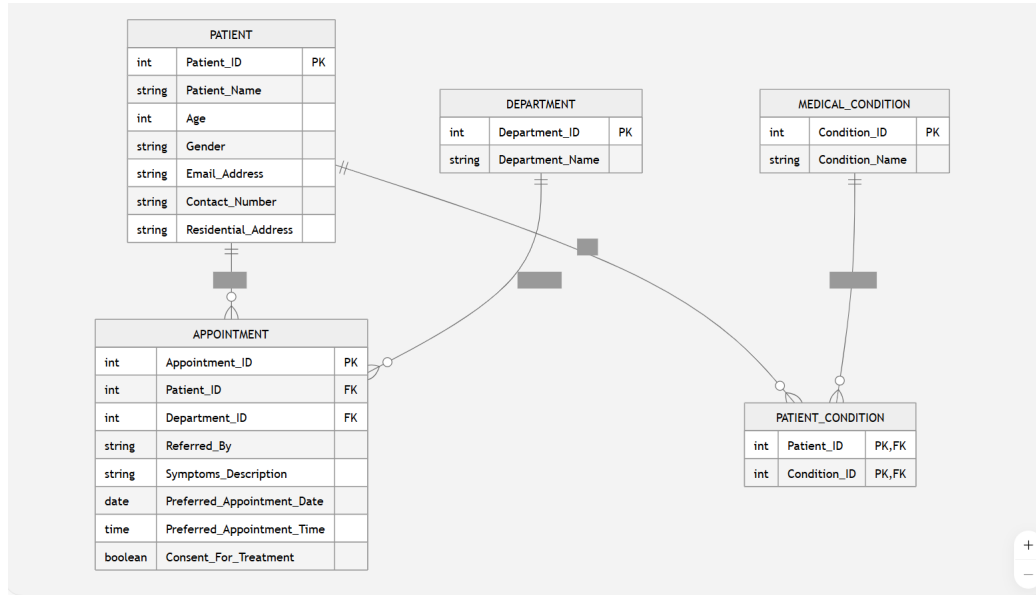


Fig. 4.1 — ER Diagram of the Appointment Booking data model

4.1 Entities and Relationships

Entity	Key Attributes	Relationship
PATIENT	Patient_ID (PK), Patient_Name, Age, Gender, Email_Address, Contact_Number, Residential_Address	One patient can have many Appointments and many Patient_Conditions
APPOINTMENT	Appointment_ID (PK), Patient_ID (FK), Department_ID (FK), Referred_By, Symptoms_Description, Preferred_Appointment_Date, Preferred_Appointment_Time, Consent_For_Treatment	Many appointments belong to one Patient and one Department
DEPARTMENT	Department_ID (PK), Department_Name	One department handles many Appointments
MEDICAL_CONDITION	Condition_ID (PK), Condition_Name	Linked to many patients through Patient_Condition

Entity	Key Attributes	Relationship
PATIENT_CONDITION	Patient_ID (PK, FK), Condition_ID (PK, FK)	Resolves the many-to-many link between Patient and Medical_Condition

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5. Form Design — Appointment_Booking_Form

Based on the ER diagram, the Appointment_Booking_Form was built in Zoho Creator's Form Builder with a total of 13 fields, grouped into four logical sections: Patient Details, Contact Details, Medical Details and Scheduling Consent. The field list, along with the field type and whether it is mandatory, is summarised below.

S.No	Field Name	Field Type	Mandatory
1	Patient Name (First / Last)	Name	Yes
2	Age	Number	Yes
3	Gender	Radio (Male/Female/Other)	Yes
4	Email Address	Email	No
5	Contact Number	Phone Number	Yes
6	Residential Address	Address	No
7	Referred By	Single Line	No
8	Department	Dropdown	Yes
9	Existing Conditions	Multi-Select Dropdown	No
10	Symptoms Description	Multi-Line	No
11	Preferred Appointment Date	Date	Yes
12	Preferred Appointment Time	Time	No
13	Consent For Treatment	Checkbox	Yes

6. Lab Deliverable Requirement

The deliverable specification for this lab exercise (shown below) required the Appointment_Booking_Form to be built exactly as specified, with the listed validations applied, an Appointment_Booking_Report generated, and a screenshot-based record submitted showing the form in Form Builder with all 13 fields, a live preview of the form, and at least three sample submitted records in the report view.

9. Deliverable for This Lab Exercise

Students must build the Appointment_Booking_Form in Zoho Creator exactly as specified above, apply the listed validations, generate the Appointment_Booking_Report, and submit a screenshot-based lab record showing: (a) the form in Form Builder with all 13 fields, (b) a live preview of the form, and (c) at least three sample submitted records in the report view.

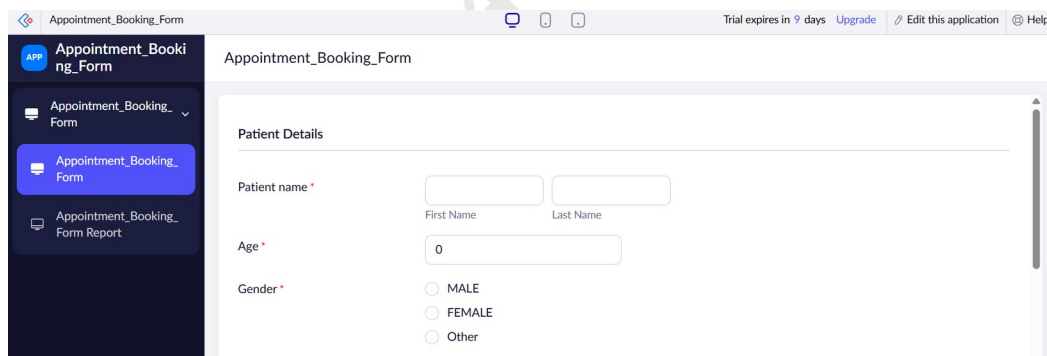
Fig. 6.1 — Deliverable requirement for the lab exercise

7. Form Builder — Live Preview Screenshots

The following screenshots show the Appointment_Booking_Form as rendered in Live Preview inside Zoho Creator, arranged section by section.

7.1 Patient Details Section

This section captures the patient's identity: Patient Name (First Name and Last Name), Age and Gender. Patient Name, Age and Gender are all marked mandatory (indicated by the red asterisk).



The screenshot displays the Zoho Creator interface for the 'Appointment_Booking_Form'. On the left, a sidebar shows the application structure with options for 'Appointment_Booking_Form' and 'Appointment_Booking_Form Report'. The main area shows the live preview of the form. The 'Patient Details' section contains three mandatory fields: 'Patient name' (with sub-fields for 'First Name' and 'Last Name'), 'Age', and 'Gender'. The 'Age' field has a value of '0' entered. The 'Gender' field has three radio button options: 'MALE', 'FEMALE', and 'Other'. The top of the interface shows the application title 'Appointment_Booking_Form' and a trial expiration notice 'Trial expires in 9 days'.

Fig. 7.1 — Patient Details section of the form

7.2 Contact Details Section

This section collects the Email Address, the mandatory Contact Number (with the +91 country code pre-selected), and the Residential Address, which is a composite field made up of Address Line 1, Address Line 2, City/District, State/Province, Postal Code and Country.

The screenshot shows a form titled "Contact Details". It contains the following fields:

- Email Address:** A single-line text input field.
- Contact Number *:** A field with a dropdown menu showing the country code "+91" and a pre-filled number "81234 56789".
- Residential Address:** A composite field consisting of:
 - Address Line 1:** A single-line text input field.
 - Address Line 2:** A single-line text input field.
 - City / District:** A single-line text input field.
 - State / Province:** A single-line text input field.
 - Postal Code:** A single-line text input field.
 - Country:** A dropdown menu with the option "-Select-" selected.

Fig. 7.2 — Contact Details section of the form

7.3 Medical Details Section

This section records the Referred By source, the mandatory Department dropdown, the Existing Conditions multi-select dropdown, and a free-text Symptoms Description field.

The screenshot shows a form titled "Medical_Details". It contains the following fields:

- Referred By:** A single-line text input field.
- Department *:** A dropdown menu with the option "-Select-" selected.
- Existing Conditions:** A multi-select dropdown menu with the option "-Select-" selected.
- Symptoms Description:** A large, multi-line text input field.

Fig. 7.3 — Medical Details section of the form

7.4 Scheduling Consent Section

The final section captures the Preferred Appointment Date (mandatory) and Preferred Appointment Time, along with the mandatory Consent For Treatment checkbox, followed by the Submit button.

Scheduling Consent

Preferred Appointment Date *

Preferred Appointment Time

Consent For Treatment * ☐ I consent to examination and treatment

Fig. 7.4 — Scheduling Consent section and Submit button

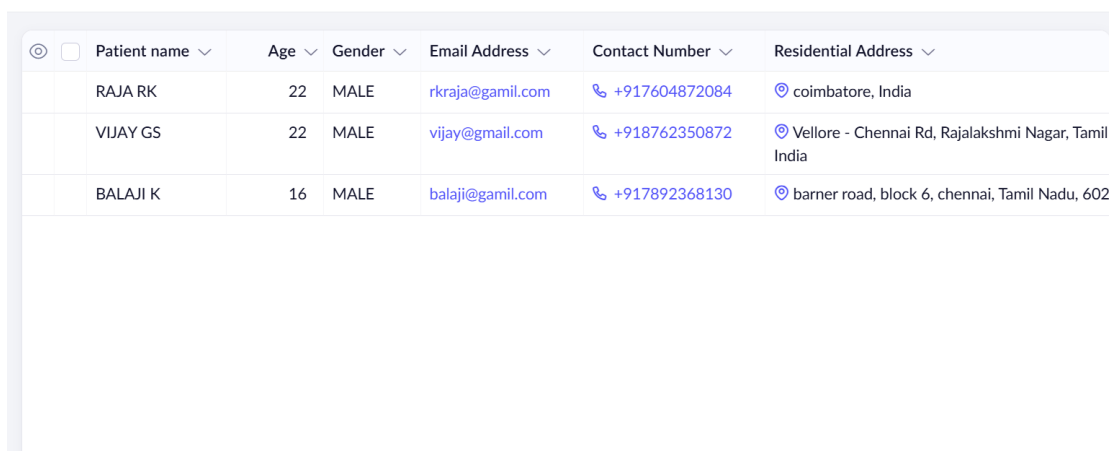
8. Validations Applied

To ensure data integrity, field-level validations were configured on the mandatory and format-sensitive fields of the form, as summarised in the table below.

Field	Validation Applied
Patient Name	Marked mandatory; must not be left blank at submission.
Age	Mandatory numeric field; only positive integer values are accepted (age of the patient).
Gender	Mandatory single-choice field restricted to Male, Female or Other.
Contact Number	Mandatory phone field with country code (+91) and a fixed 10-digit pattern for Indian mobile numbers.
Email Address	Field-level email format check (must contain a valid '@' pattern) when a value is entered.
Department	Mandatory dropdown restricted to the pre-configured list of hospital departments.
Preferred Appointment Date	Mandatory date field; past dates are restricted so patients can only book present or future dates.
Consent For Treatment	Mandatory checkbox — the form cannot be submitted unless the patient consents to examination and treatment.

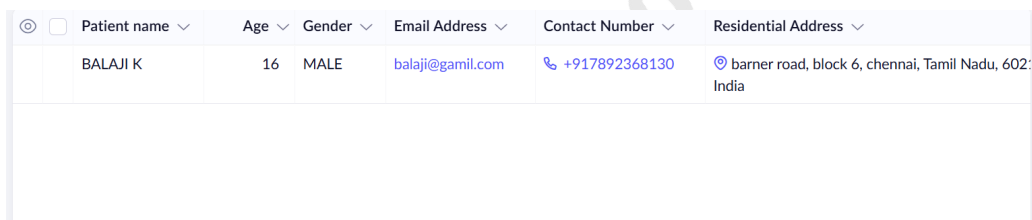
9. Appointment_Booking_Report — Submitted Records

After publishing the form, three sample records were submitted to test the form end-to-end. The submitted data is automatically listed in the Appointment_Booking_Form Report view, which can be scrolled horizontally to see every field of every record. The screenshots below show the report view for the submitted records.



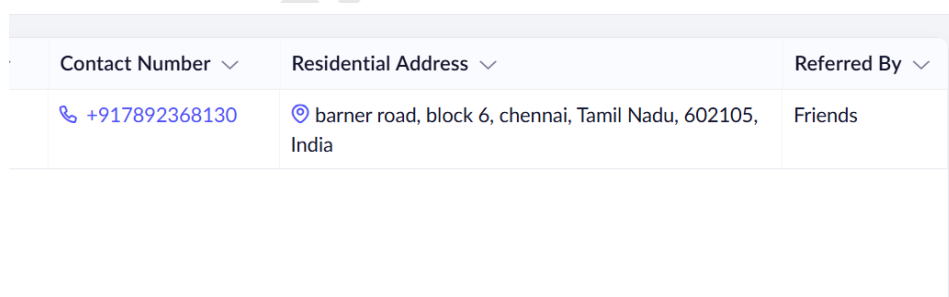
	☐ Patient name ▾	Age ▾	Gender ▾	Email Address ▾	Contact Number ▾	Residential Address ▾
	RAJA RK	22	MALE	rkraja@gamil.com	+917604872084	coimbatore, India
	VIJAY GS	22	MALE	vijay@gmail.com	+918762350872	Vellore - Chennai Rd, Rajalakshmi Nagar, Tamil India
	BALAJI K	16	MALE	balaji@gamil.com	+917892368130	barner road, block 6, chennai, Tamil Nadu, 602

Fig. 9.1 — Report view showing three submitted sample records (Patient Name, Age, Gender, Email, Contact Number, Residential Address)



	☐ Patient name ▾	Age ▾	Gender ▾	Email Address ▾	Contact Number ▾	Residential Address ▾
	BALAJI K	16	MALE	balaji@gamil.com	+917892368130	barner road, block 6, chennai, Tamil Nadu, 602: India

Fig. 9.2 — Report view highlighting a single submitted record (BALAJI K)



Contact Number ▾	Residential Address ▾	Referred By ▾
+917892368130	barner road, block 6, chennai, Tamil Nadu, 602105, India	Friends

Fig. 9.3 — Report view showing the Contact Number, Residential Address and Referred By columns

9.1 Summary of Submitted Sample Records

Patient Name	Age	Gender	Email Address	Contact Number
RAJA RK	22	MALE	rkraja@gamil.com	+91 76048 72084
VIJAY GS	22	MALE	vijay@gmail.com	+91 87623 50872
BALAJI K	16	MALE	balaji@gamil.com	+91 78923 68130

10. Result

The Appointment_Booking_Form was successfully data-modelled and built in Zoho Creator with all 13 required fields spread across four logical sections (Patient Details, Contact Details, Medical Details and Scheduling Consent). The specified field-level validations — including mandatory fields, the numeric Age field, the formatted Contact Number, and the mandatory Consent For Treatment checkbox — were applied and verified. The Appointment_Booking_Report was generated, and three sample records were submitted and verified to appear correctly in the report view, confirming that the form design and data model satisfy the requirements of the lab exercise.